

Barangay Health Center System - User Guide

Integrated Patient Information System with Queue Management for Barangay Health Centers.

Gawad BIS integration: Resident registration, SSO, staff import, and medicine sync are covered in [INTEGRATED_USER_GUIDE.md](#).

Exporting to PDF: This file uses plain ASCII punctuation (hyphens, quotes) and is saved as **UTF-8 with BOM** for Windows tools. In Word: *File -> Open -> select this .md file, then File -> Save As -> PDF*. In VS Code: install *Markdown PDF* extension, then export. If characters still look wrong, ensure your converter encoding is set to **UTF-8**.

1. System overview

The BHC System helps health center staff:

- Register and manage **patient records** (linked to Gawad BIS when integration is enabled)
- Route patients to the correct **service station** based on reason for visit
- Run **multi-station queues** (Registration -> Triage -> Consultation -> Pharmacy)
- Capture **triage vitals**, **consultation records**, and **medicine dispensing**
- Schedule **follow-up appointments**
- Issue **clinical documents** (prescription receipts, medical certificates, referrals)
- Show **"Now serving"** on a TV/display for the waiting area
- Give patients a **paperless ticket** via QR code on their phone
- Review **reports** and **audit logs** (admin)

Default stations

Station	ID	Purpose
Registration	1	Routing desk only - creates tickets for other stations
Triage / Vitals	2	Initial assessment and vitals
Consultation	3	Doctor/nurse consultation
Pharmacy	4	Medicine dispensing

Who operates the queue?

The queue is **staff-operated**. Staff call patients when ready. The system can **auto-call the next patient** after Complete or Skip to reduce manual clicking during busy hours.

Doctors use **My patients** for their assigned consultation queue; they do not operate Registration, Triage, or Pharmacy desk pages.

2. Accessing the system

Staff pages (login required)

Use your health center's server address. Examples:

Setup	Login URL
XAMPP (recommended)	http://YOUR_LAN_IP/bhc_system/public/login
XAMPP (no URL rewrite)	http://YOUR_LAN_IP/bhc_system/public/index.php/login
PHP dev server	http://YOUR_LAN_IP:8000/login

Replace **YOUR_LAN_IP** with the computer's IPv4 address (e.g. **192.168.1.49**).

You may also arrive signed in via **Gawad BIS** (SSO) when clicking a Health Center link — see the integration guide.

Patient-facing pages (no login)

Page	URL pattern	Purpose
Station display	/display/{stationId}	TV screen: Now serving + Next
Ticket status	/ticket/{ticketId}	Patient's live ticket on phone
Fullscreen QR	/ticket/{ticketId}/qr	Large QR for printing/showing

Example (XAMPP): http://192.168.1.49/bhc_system/public/display/2 (Triage)

Demo login (first install)

If no users exist in the database, a default admin is created automatically:

- **Username:** **admin**
- **Password:** **admin123**

Change this password before real deployment.

3. User roles

Role	Access
Admin	All staff features + Reports + Activity Log + Staff accounts + Medicine list + Gawad staff import
Staff	Staff dashboard, Patients, Appointments, Health Center queues (Registration through Pharmacy)
Doctor	My patients (/doctor) — assigned consultation queue, patient charts, consultations, clinical documents

After login

Role	Lands on	Sidebar / menu
Staff	Staff dashboard (/staff)	Patients, Appointments, Health Center (queue submenu)
Admin	Admin dashboard (/admin)	Same as staff + Administration (Reports, Activity Log, Medicine list, Staff accounts)
Doctor	My patients (/doctor)	Doctor menu only

Staff and doctors cannot open admin pages (reports, audit, staff accounts) by URL. Doctors cannot open clinic desk queue pages.

Account menu (top-right, all roles)

Link	Purpose
Change password	Update your own login password
Name on documents	Doctors only — how your name appears on certificates and referrals

Create a staff account (admin)

Administration -> Staff accounts (or /users).

1. Click **Add staff account** (or **Import Gawad staff** when syncing from Gawad BIS).
2. For manual create: enter **username**, **password** (min. 8 characters), optional **display name**, and **role** (Staff, Doctor, or Admin).
3. Click **Create account**.

Action	Use when
Import Gawad staff	Pull barangay users from Gawad BIS for SSO (see integration guide)
Reset password	Staff forgot password
Deactivate	Person left or should not sign in (cannot deactivate your own account)
Activate	Re-enable a deactivated account

At least one **active admin** must remain; the system will not deactivate the last admin.

Doctors are created manually in BHC with role **Doctor** (not auto-imported from Gawad).

Create a staff account (command line — optional)

From the project folder:

```
php scripts/create_staff_user.php nurse1 YourPassword staff
php scripts/create_staff_user.php dr.cruz YourPassword doctor
```

4. Navigation (after login)

Main sidebar

Menu	Description
Staff / Admin dashboard	Daily workspace (/staff or /admin)
My patients	Doctor dashboard only (/doctor)
Patients	Patient list, registration, history
Appointments	Upcoming follow-up visits (/appointments)
Health Center	Expandable submenu (see below)
Reports	Report hub (admin only)
Activity Log	Audit trail (admin only)
Medicine list	Local fallback names; shows Gawad sync status (admin only)
Staff accounts	User management (admin only)
Logout	End session

Health Center submenu

Link	Path	Purpose
Registration Queue	/queue/1	Patient routing desk
Triage / Vitals	/queue/2	Triage station queue
Consultation	/queue/3	Consultation desk queue
Pharmacy Queue	/queue/4	Pharmacy station queue
All Stations	/stations	Station list overview
Queue Management	/coordinator	All stations on one screen

Before login, only a simple header (Home + Login) is shown.

5. Daily workflow (step by step)

Step 1 - Register or find a patient

Staff (standard path):

1. Register the resident in **Gawad BIS**, then use **Register at Health Center** on the resident profile (prefills BHC form via SSO).
2. Or use **Emergency walk-in** on the Patients page when immediate care is required and Gawad registration is not possible first.

Admin:

- May use **Add patient** directly, or follow the Gawad path above.

On the registration form:

1. Enter **first name**, **last name**, **sex**, and **date of birth** (required).
2. Optional: middle name, suffix, contact, address, PhilHealth, emergency contact, notes.
3. **Confirm Balong Bato residency** (required for new registrations).
4. The system assigns a **BHC ID** (e.g. **BHC-000001**).
5. While typing the name, **duplicate suggestions** appear — click a match to use an existing patient instead of creating a duplicate.

See [INTEGRATED_USER_GUIDE.md](#) for Gawad handoff, linking, and emergency walk-in rules.

Step 2 - Route at Registration (routing desk)

1. Open **Health Center -> Registration Queue** (Station 1), or **Patient Routing** from the staff dashboard.
2. **Search** the patient by name or BHC ID.
3. Enter **Reason for visit** (free text; suggestions may appear).
4. Choose **Assign / route to station** (Triage, Consultation, or Pharmacy).
5. Click **Create ticket**.

If the patient has an **appointment today**, a banner appears on the routing form.

The patient receives a **ticket number** (e.g. **TR-001** for Triage) and can open the **ticket screen** with QR code.

Note: Only **one consultation ticket per patient per day** is allowed. The system warns if you try to route twice to Consultation.

Step 3 - Station staff serves the queue

For each service station (Triage, Consultation, Pharmacy):

1. Open the station from **Health Center**, or use **Queue Management**.
2. **Call next** or **Call** a specific waiting ticket.
3. When a patient is **Now serving**, complete the clinical form for that station (see **Section 6**).
4. **Complete** or **Skip** when done at this station.

Recall to queue: If a skipped patient returns the same day at the same station, use **Recall** on that ticket (station page or Queue Management). The same ticket number returns to **waiting**.

Auto-call: After **Complete** or **Skip**, the system automatically calls the **next waiting** patient if the station is idle. **Recall** does not auto-call.

Consultation — assign doctor: At the Consultation desk, when a patient is Now serving, staff can **assign a doctor** from the dropdown. The patient then appears on that doctor's **My patients** queue.

Step 4 - Patient knows their turn

Patients can be informed in three ways:

1. **TV / monitor - Patient display** for that station (`/display/{stationId}`). See **Section 16** for setup. Tap **Enable sound** once; the display **flashes** and plays a **chime** when Now serving changes.
2. **Phone (QR ticket)** - Patient scans QR on ticket screen. Tap **Enable sound** on the ticket page. Page refreshes every 5 seconds; when status becomes **SERVING**, the patient sees "*You are being called now*" and hears a **chime**.
3. **Waiting area** - Staff may still call names verbally; display and ticket are backups.

Mobile push notifications (SMS / app alerts) are not included.

Step 5 - End of visit

- Mark ticket **Complete** at the last station (usually Pharmacy).
- If the patient needs another station, create a **new routed ticket** from Registration.
- Schedule a **follow-up appointment** from the patient's history page if needed.

6. Clinical workflow at stations

When a ticket is **Now serving**, station staff capture clinical data before completing the ticket.

Triage / Vitals (Station 2)

Open **Health Center** -> **Triage / Vitals**, call the patient, then enter:

Field	Examples
Blood pressure	Systolic / diastolic (e.g. 120 / 80)
Temperature	e.g. 36.5 C
Pulse rate	e.g. 72
Weight / Height	kg and cm
Triage notes	Chief complaint, observations

Click **Save triage record**, then **Complete** when the patient may proceed to the next station.

Consultation (Station 3 — desk)

Open **Health Center** -> **Consultation**. When Now serving:

1. Optionally **assign a doctor** (patient appears on doctor's **My patients**).
2. Enter **diagnosis** (required) and **clinical notes**.
3. Optionally add **prescribed medicines** (picker shows Gawad inventory stock when sync is enabled).
4. Click **Save consultation record**, then **Complete**.

Doctors may instead work from **My patients** (Section 7) for the same consultation forms and documents.

If the patient had an **appointment today**, saving the consultation can mark that appointment completed.

Pharmacy (Station 4)

Open **Health Center** -> **Pharmacy Queue**. When Now serving:

1. Review medicines from the consultation record (or enter dispensed items).
2. Record quantities and source (**Clinic stock**, **Request from LGU**, or **Buy externally**).
3. Click **Record medicines dispensed**, then **Complete**.

Stock hints come from **Gawad BIS** when medicine sync is active. Update barangay inventory in Gawad separately after dispensing.

7. Doctor portal

Doctors sign in and land on **My patients** (</doctor>).

Assigned patient queue

The dashboard shows patients **assigned to you** at Consultation today (waiting and serving combined):

Action	When
Call next	Call the oldest waiting patient assigned to you
Call	Call a specific waiting patient
Complete	Consultation finished
Skip	Patient absent

The page **auto-refreshes every 5 seconds** when your queue changes. Tap **Enable sound** for a chime when new patients are waiting. If you are editing a form, a banner offers **Refresh now** instead of interrupting you.

Patient clinical record

Click a patient to open their chart (</doctor/patients/{id}>):

- Demographics and registry notes
- Today's vitals and past triage
- Consultation history
- Add **diagnosis**, **clinical notes**, and **medicines**
- Issue **medical certificate**, **referral**, or **recommendation**
- View issued **clinical documents**

When the patient is **Now serving**, save the consultation record, then **Complete consultation**.

Name on documents

Account menu (top-right) -> **Name on documents** — set how your name prints on certificates and referrals (e.g. *Dr. Maria S. Cruz, M.D.*).

8. Queue Management (busy days)

Queue Management (</coordinator>) is a single dashboard for all service stations.

Use it when:

- One person monitors multiple stations
- You want fewer clicks than opening each station separately

For each station you see:

- Waiting / Serving counts
- **Now serving** with Complete / Skip
- **Call next** when idle
- **Next waiting** list with individual **Call** buttons
- **Skipped today** list with **Recall**

The page **auto-refreshes every 5 seconds**. Auto-call rules are the same as on individual station pages.

9. Patients module

Action	How
View all patients	Patients -> browse list (paginated, searchable)
Register (staff)	Register via Gawad BIS or Emergency walk-in
Register (admin)	Add patient or Gawad path
Edit patient	Patients -> select patient -> Edit
Patient history	Patients -> select patient -> History (visits, triage, consultations, medicines, documents)
Schedule appointment	From patient History page
Search while routing	Registration desk search field
Archive / restore	Admin actions on patient record

10. Appointments

Appointments ([/appointments](#)) lists scheduled follow-up visits across all patients.

Feature	Description
Date filter	Show appointments from/to selected dates
Status	Scheduled, completed, cancelled, no-show
Patient link	Click patient name to open history
At Registration	Banner when routing a patient with an appointment today
At Consultation	Saving consultation can complete today's linked appointment

Create appointments from a patient's **History** page when scheduling a return visit.

11. Reports (admin)

Open **Reports** (</reports>) for the report hub. Each report supports date filtering and **CSV export**.

Report	Path	Content
Daily operations	/reports/daily	Visits, queue tickets by station, triage counts, reasons, hourly arrivals
Queue report	/reports/monthly	Tickets per station, completed/skipped, average wait and service times
Clinical summary	/reports/clinical	Consultations, diagnoses, medicines prescribed/dispensed, receipts
Appointments	/reports/appointments	Scheduled, completed, cancelled, no-show by period

Queue reports use ticket **creation date** in the selected period. Completed tickets remain in the database for history.

12. Activity Log (admin)

Activity Log (</audit>) shows who did what and when, including:

Action	When logged
login / logout / login_failed	Sign in, sign out, wrong password
patient_create / patient_update	Patient registered or edited
ticket_create , ticket_call , ticket_call_next , ticket_call_next_auto	Queue actions
ticket_complete , ticket_skip , ticket_recall	Ticket finished, skipped, or recalled
user_create , user_deactivate , user_activate , user_password_reset	Staff account changes
user_password_change	Self-service password change
gawad_staff_sync , gawad_link	Integration events (when used)

Useful for accountability and incident review.

13. Medicine list (admin)

Medicine list (</medicines>) manages the **local fallback** catalog (medicine names for the prescription picker).

When **Gawad medicine sync** is enabled:

- Consultation and pharmacy pickers use the **live Gawad BIS inventory** (names + stock levels).
- This page shows sync status and links to **Open Gawad BIS Medicines** for stock management.

When sync is unavailable, pickers fall back to the local list on this page.

See [INTEGRATED_USER_GUIDE.md](#) Section 9 for medicine sync details.

14. Ticket statuses

Status	Meaning
waiting	In queue, not yet called
serving	Currently being served (Now serving)
done	Completed successfully
skipped	Patient absent or could not be served (can Recall to queue same day at same station)

Queues are **daily** (based on ticket creation date). Old tickets stay in the database for reports. **Recall** only works for **today's** skipped tickets at that station.

15. QR codes and LAN setup

For patients to scan QR codes on their phones, phones must be on the **same Wi-Fi/LAN** as the server.

Edit `config/app.local.php` (or `config/app.php`):

```
'base_url' => 'http://YOUR_LAN_IP/bhc_system/public',
```

Examples:

- XAMPP: `http://192.168.1.49/bhc_system/public`
- With index.php fallback: `http://192.168.1.49/bhc_system/public/index.php`

Do **not** use `localhost` in `base_url` if patients use phones — they cannot reach your PC's localhost.

16. Waiting area display setup (TV / monitor / tablet)

The **Patient display** page shows **Now serving**, the **Next** waiting tickets, a **flash** when the number changes, and a **chime** when staff call the next patient. Tap **Enable sound** once per browser (required by Chrome/tablets). It refreshes automatically every **3 seconds**. **No login** is required on the TV or tablet.

Patient ticket page (`/ticket/{id}`): same **Enable sound** + chime when that ticket becomes **SERVING** (for patients watching their phone).

What you need

Item	Notes
Server PC	XAMPP running; fixed LAN IP (e.g. 192.168.1.49)
Display device	TV + mini PC, wall tablet, or second monitor
Network	Display device on the same Wi-Fi/LAN as the server
Browser	Chrome or Edge (recommended)

Display URLs (bookmark these)

Replace **YOUR_LAN_IP** with your server's IPv4 address.

Waiting area	Station ID	Bookmark URL (XAMPP)
Triage / Vitals	2	http://YOUR_LAN_IP/bhc_system/public/display/2
Consultation	3	http://YOUR_LAN_IP/bhc_system/public/display/3
Pharmacy	4	http://YOUR_LAN_IP/bhc_system/public/display/4

If clean URLs do not work, use:

http://YOUR_LAN_IP/bhc_system/public/index.php/display/2 (change 2 to 3 or 4 as needed)

Do not use Station 1 (Registration) for a waiting-area screen. Registration is the routing desk only; patient queues run at Triage, Consultation, and Pharmacy.

Open from the system: Health Center -> choose a station -> **Patient display**, or use display links on the dashboard.

Hardware options

Option	Best for
TV + mini PC / old laptop (HDMI)	Main waiting room; large numbers easy to read
Wall tablet (Android / iPad + mount)	Small room; use ticket numbers only (see privacy below)
Second monitor at the station	Low cost; staff see queue + display side by side

Setup steps (first time)

1. **Server:** Start XAMPP (Apache + MySQL). Note the PC's LAN IP (e.g. **192.168.1.49**).
2. **Test:** On a phone or the display device, open http://YOUR_LAN_IP/bhc_system/public/display/2. The page must load (not **localhost**).
3. **Display device:** Open the correct URL for that waiting area (Triage = **2**, Consult = **3**, Pharmacy = **4**).
4. Click **Fullscreen** on the page (or press **F11**).
5. **Sound:** Tap **Enable sound** on the display (or ticket page on a phone); turn up TV/tablet/phone volume. You should hear a short test chime.
6. **Privacy:** Leave **Show names** off unless your center approves names on a public screen. Default shows **ticket numbers only** (e.g. **TR-003**).

7. **Power:** Set the display device so the screen does **not** sleep while plugged in.
8. **Leave the tab open** all day; it auto-refreshes - do not close it during clinic hours.

Windows kiosk mode (optional - auto-open on startup)

Create a desktop shortcut to Chrome (edit the IP and station ID):

```
"C:\Program Files\Google\Chrome\Application\chrome.exe" --kiosk --noerrdialogs
"http://192.168.1.49/bhc_system/public/display/2"
```

Put the shortcut in the **Startup** folder so the display opens after reboot.

How many displays?

Layout	Suggestion
One shared waiting room (Triage only)	1 TV -> /display/2
Separate consult waiting	Add 1 TV -> /display/3
Pharmacy counter area	Add 1 TV -> /display/4
Very small center	Start with one display at the busiest station

Staff still **Call next** / **Complete** from **Health Center** or **Queue Management**; the TV updates within a few seconds.

Daily routine

When	Action
Morning	Start server (XAMPP) -> open display URL -> Enable sound -> Fullscreen -> test chime
During clinic	Keep display tab open; staff operate queues as usual
Evening	Close browser optional; server can stay on for reports

Display checklist

- ☐ Server has static LAN IP
- ☐ Display opens correct station URL ([/display/2](#), [3](#), or [4](#))
- ☐ Same Wi-Fi/LAN as server
- ☐ Fullscreen or kiosk mode enabled
- ☐ Chime tested (one click on page first)
- ☐ Staff trained: Call next / Complete updates the screen

17. Quick reference - URLs

Page	Path
Public home	/
Staff dashboard	/staff
Admin dashboard	/admin
Doctor — My patients	/doctor
Login	/login
Patients	/patients
Add patient	/patients/create
Patient history	/patients/{id}/history
Edit patient	/patients/{id}/edit
Appointments	/appointments
Change your password	/account/password
Name on documents (doctor)	/account/document-name
All stations	/stations
Registration routing	/queue/1
Triage queue	/queue/2
Consultation queue	/queue/3
Pharmacy queue	/queue/4
Queue Management	/coordinator
Reports hub	/reports
Daily report	/reports/daily
Queue report	/reports/monthly
Clinical report	/reports/clinical
Appointments report	/reports/appointments
Activity Log	/audit
Medicine list	/medicines
Staff accounts	/users
Patient display	/display/{stationId}
Patient ticket	/ticket/{ticketId}

Station IDs (default): 1 = Registration, 2 = Triage, 3 = Consultation, 4 = Pharmacy.

18. Troubleshooting

Problem	What to check
Login redirects to wrong URL	Use full path: <code>/bhc_system/public/login</code> not <code>/login/login.php</code>
QR does not open on phone	Set <code>base_url</code> to LAN IP; same Wi-Fi; allow firewall
Search patient not working	Hard refresh (Ctrl+F5); check browser console
Sidebar missing	Sidebar appears only after login
Auto-call did not happen	No more waiting patients; or station still has someone serving
404 on clean URLs	Use <code>.../public/index.php/login</code> or enable Apache <code>mod_rewrite</code>
Display page blank or cannot connect	Use LAN IP, not <code>localhost</code> ; check Apache and Windows Firewall
Display never changes	Staff must Call next ; page auto-refreshes every 3 seconds
No chime on TV	Click Fullscreen or the page once; check volume; use Chrome/Edge
Wrong ticket numbers on screen	Wrong bookmark (e.g. Pharmacy TV must use <code>/display/4</code>)
Display asks for login	Use <code>/display/2</code> not <code>/queue/2</code> (staff queue page)
Staff cannot open Add patient	By design — use Gawad or Emergency walk-in (integration guide)
Doctor blocked from queue desk	Doctors use My patients , not Consultation queue page
Second consultation ticket rejected	One consultation ticket per patient per day
Medicines missing in picker	Gawad sync off or unavailable — see Medicine list (admin)
SSO / Gawad registration issues	See INTEGRATED_USER_GUIDE.md

19. Recommended setup at the health center

1. **Registration PC** — routing desk; Gawad BIS for resident enrollment
2. **One PC per station** (or shared via Queue Management) — Triage, Consult, Pharmacy
3. **Doctor PC** — **My patients** and patient charts
4. **TV/monitor per waiting area** — see **Section 16** for display URLs
5. **One server PC** — XAMPP + Gawad BIS + MongoDB on LAN; set static IP
6. **Admin account** — staff logins, reports, Gawad staff import

20. Workflow diagram

Interactive Mermaid diagrams (recommended for editing and high-quality export): see [WORKFLOW_DIAGRAM.md](#) — copy into [Mermaid Live Editor](#) to preview and export PNG/SVG.

Hand-drawn / exported workflow diagram: see [WORKFLOW_DIAGRAM.md](#) to generate PNG or SVG from the Mermaid sources ([workflow-diagram.mmd](#), [ticket-lifecycle.mmd](#)).

21. Related documents

Document	Content
README.md	Documentation index
INTEGRATED_USER_GUIDE.md	Gawad BIS integration (SSO, resident handoff, staff import, medicine sync)
SOURCE_CODE_AND_DOCUMENTATION.md	Technical architecture and APIs
WORKFLOW_SUMMARY.md	Process overview and ticket lifecycle
WORKFLOW_DIAGRAM.md	Visual workflow diagrams
DEPLOY_INFINITYFREE.md	Cloud deployment guide

Document version: BHC System with staff/admin/doctor roles, Health Center queue menu, triage/consultation/pharmacy clinical workflow, doctor portal, appointments, reports hub, Gawad medicine sync, waiting-area displays, Queue Management, auto-call, and Activity Log.